

CS305 Lab 4

Advanced HTTP & Web Programming

Dept. Computer Science and Engineering
Southern University of Science and Technology

Part A.

Advanced HTTP

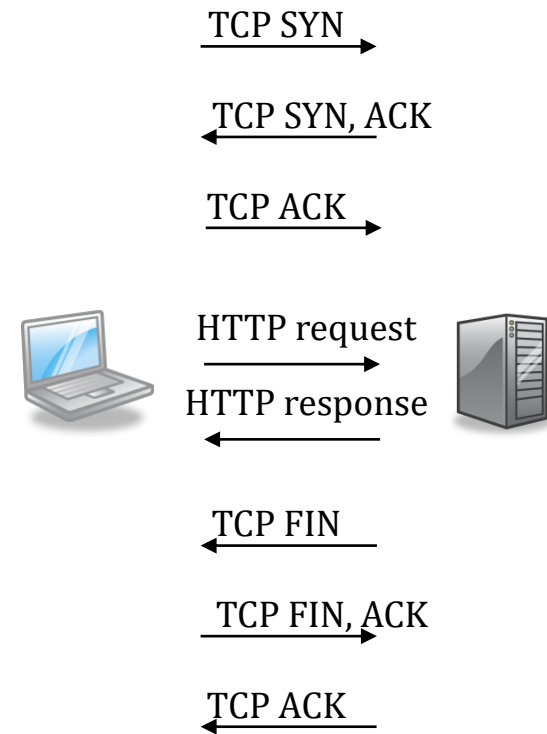
Part A.1

Connection and transfer encoding

- Connection management
 - Persistent connection, parallel connection
 - Connection: close
- Content-Length vs. Chunked transfer encoding
 - Reducing latency of response

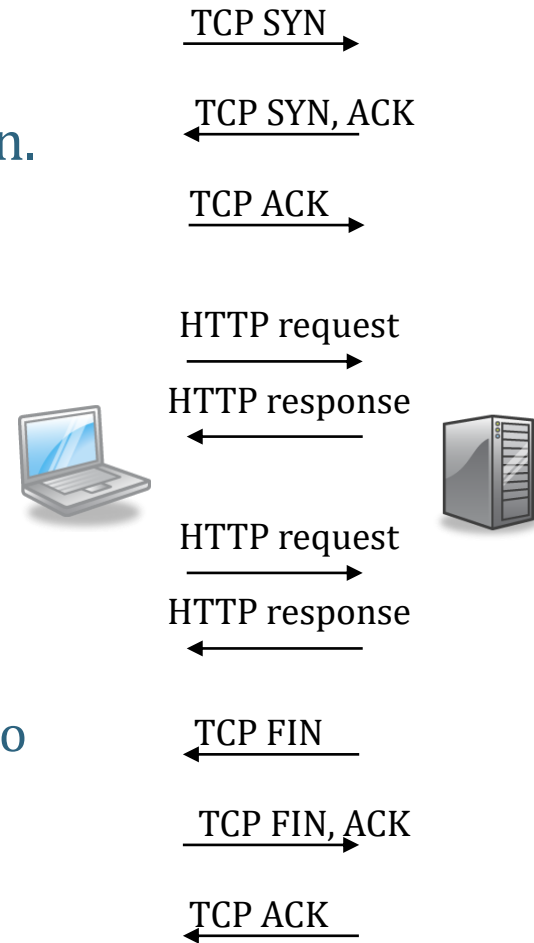
Problem in HTTP/1.0 connection

- HTTP/1.0 uses a new connection for each HTTP transaction.
- Making TCP connection slow.
 - takes three packets to establish
 - takes three packets to close
- A web page typically contains many embedded images. HTTP/1.0 would make many TCP connections to load a web page.
 - Slow page loading



Persistent and Parallel Connection

- **Persistent connection:** multiple requests and responses are sent through one TCP connection.
 - Default in HTTP/1.1
 - Browsers keep TCP connection after page load.
Why?
- **Parallel connection:** A web browser opens several TCP connections to a web site and downloads components of a web page concurrently.
 - using tcp.srcport and tcp.dstport in Wireshark to trace a http session

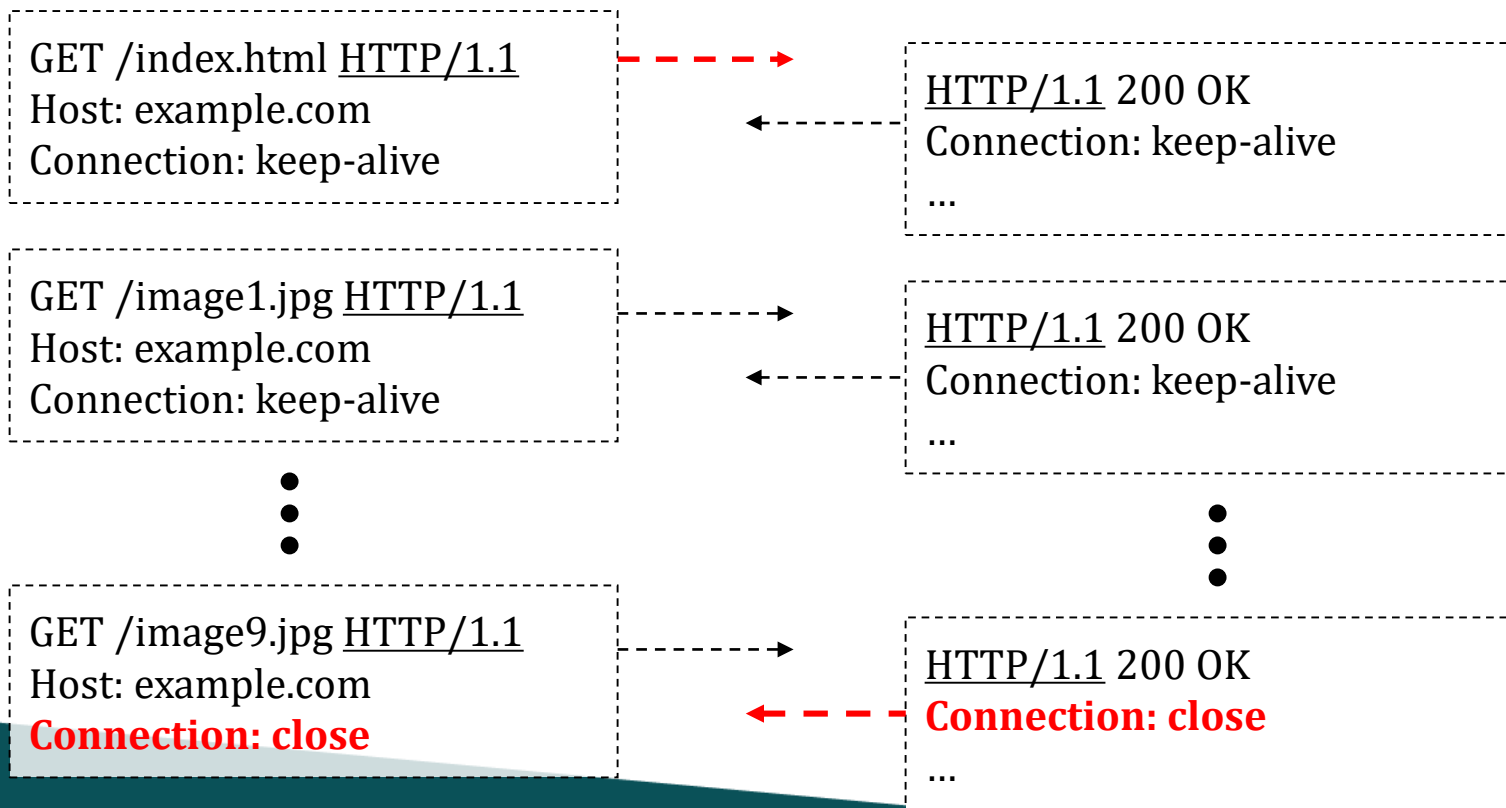


Connection:

- The Connection: header indicates whether to keep or close the current connection.
- **Connection: keep-alive.** Default, may be omitted.
- **Connection: close**
 - In a request, the client asks the server to close the connection after sending the response
 - In a response, the server indicates that it will close the connection after sending this response

Persistent connection

The client creates a connection before sending the first request. Subsequent requests and responses are transferred in this TCP connection.



Multiple messages in a connection

- A browser receives multiple responses in a TCP connection.
- To break the byte stream into messages, it must know the end of each message.
- One solution is that the server declares **Content-Length** for each response.

HTTP/1.1 200 OK

Content-Type: ...

Content-Length: 100

... content of first resource ...

... 100 bytes ...

HTTP/1.1 200 OK

Content-Type: ...

Content-Length: 200

... content of second resource ...

... 200 bytes ...

HTTP/1.1 200 OK

Connection: close

Content-Type: ...

Content-Length: 120

... content of third resource ...

... 120 bytes ...

Problems of Content-Length

- **Content-Length** header is sent before the message body.
- If a resource is generated dynamically by a server-side script (e.g. ASPX, PHP), the web server can determine Content-Length only after the script finishes execution.
 - The server first has to buffer the whole response before it can start sending the response.
- Efficiency problems:
 - Larger memory overhead
 - Slower response time

Chunked Transfer-Encoding

- **Chunked transfer-encoding** enables a web server to start transmitting the beginning parts of a response while it is still generating the rest.
 - Does not send **Content-Length**. Sends **Transfer-Encoding: chunked** instead.
 - A long response body is divided into several pieces called chunks.
 - Before sending each chunk, the server sends its length in hexadecimal.
 - After sending the last chunk, the server sends a 0.

Response with Content-Length

HTTP/1.1 200 OK

Date: Wed, 19 Mar 2008 01:46:57 GMT

Content-Type: text/plain

Content-Length: 42

abcdefghijklmnopqrstuvwxyz1234567890abcdef

Response with chunked body

HTTP/1.1 200 OK

Date: Wed, 19 Mar 2008 01:46:57 GMT

Content-Type: text/plain

Transfer-Encoding: chunked

1a

abcdefghijklmnopqrstuvwxyz

10

1234567890abcdef

0

Length of a chunk in Hex

Length of a chunk in Hex

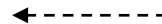
*Length of 0 means no more
chunks*

Partial Content

- How to retrieve a slice of resource?
 - Request:
 - Range: <unit>=<range-start>-<range-end>
e.g. Range: bytes=200-1000, 2000-6576, 19000-
 - Response:
 - HTTP/1.1 206 Partial Content
 - Content-Range: <unit> <range-start>-<range-end>/<size>
e.g. Content-Range: bytes 21010-47021/47022
- References:
 - <https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Range>
 - <https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Content-Range>
 - <https://developer.mozilla.org/en-US/docs/Web/HTTP/Status/206>

Example:

GET /video.mp4 HTTP/1.1
Host: example.com
Connection: keep-alive
Range: bytes=1900-2900



HTTP/1.1 **206 Partial Content**
Connection: keep-alive
Content-Range: bytes 1900-2900/4702
...

Part A.2

State, session and security

- Session management
 - HTTP is stateless
 - Cookies
 - As in common web framework (e.g. asp.net, php)
 - Session hijacking
- Encryption and SSL
 - Secured login vs. full-session HTTPS
 - Partially secure web page

HTTP is stateless

- Statelessness means that every HTTP request happens in complete isolation.
- When the client makes a HTTP request, it includes all information necessary for the server to fulfill that request.
- The server never relies on information from previous requests. If that information was important, the client would have sent it again in this request.
 - A web server does not retain info between processing of requests from a user session
 - The client and server do not need to maintain a common state

(from O'Reilly RESTful web service)

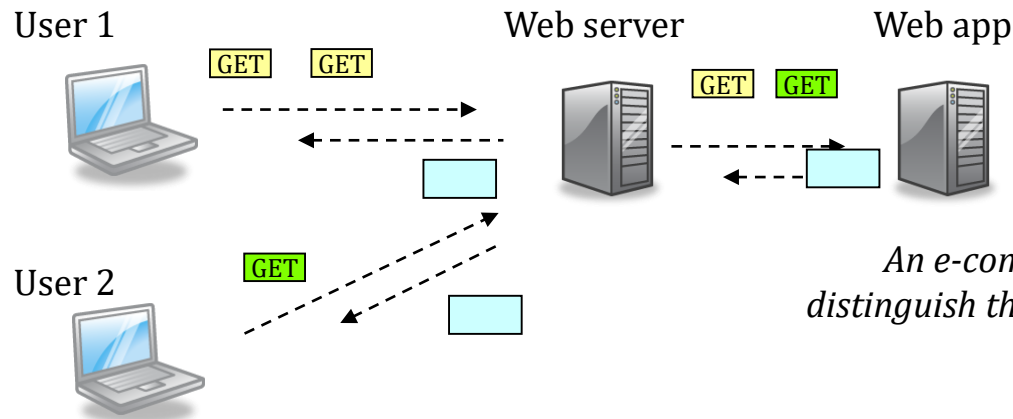
Example

- A browser keeps sending same (or similar) headers to a web server in a series of requests, e.g.
 - Host
 - User-agent
 - Content negotiation
- Each request contain the information from the full URL (absolute URL).

```
GET /wiki/Internet HTTP/1.1
Host: en.wikipedia.org
User-Agent: Mozilla/5.0 ... Firefox/3.5.3
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8
Accept-Language: en-us,en;q=0.5
Accept-Charset: ISO-8859-1,utf-8;q=0.7,*;q=0.7
```


Session Management in Web App

- Although HTTP is stateless, web app needs to maintain states in processing requests from a user session.
 - e.g. Has a user logged in?
Which requests come from the user?
- The client needs to attach session identifier in each request

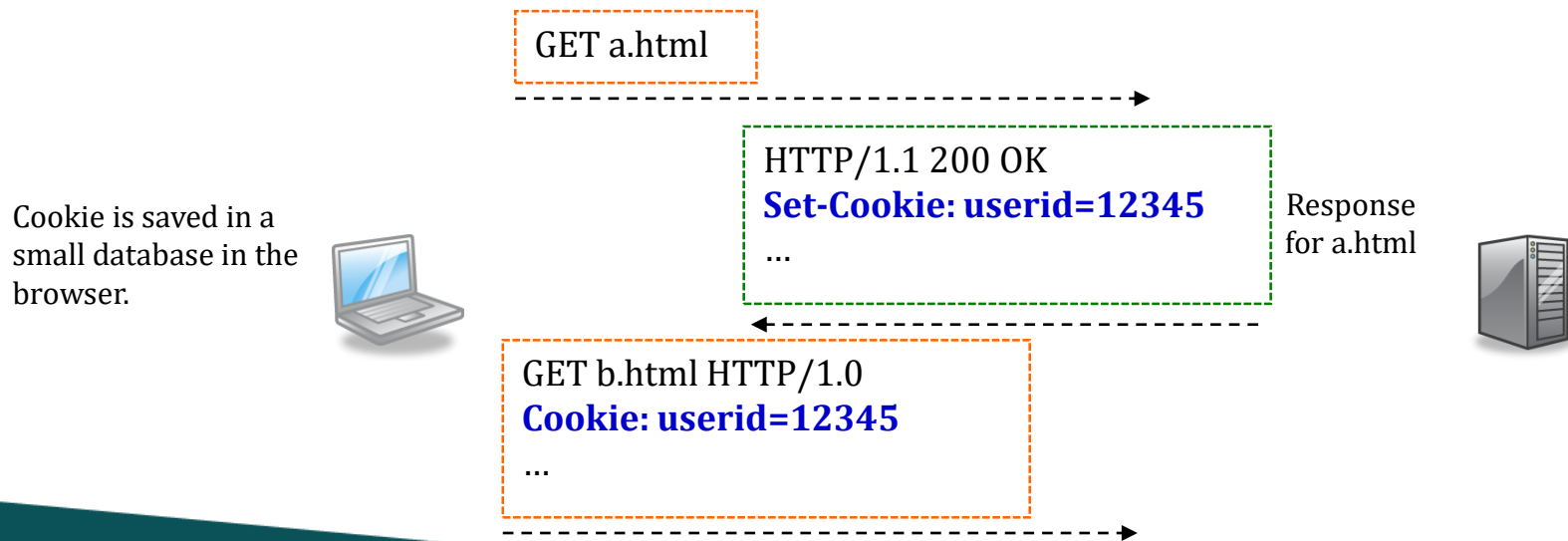


Session Management

- A web app has to track a user's progress from one request to another.
- Each request has to include some data to identify a user session.
- Common approaches:
 - Cookies
 - Hidden form field (<input type="hidden">)
 - Query string

HTTP Cookies

- Cookies are small pieces of data a web server asks a client to keep and send back in future requests.
 - Servers add header **Set-Cookie: name=value** in response
 - Clients add header **Cookie: name=value** in future requests



Example: Typical use of Session id



POST login.php HTTP/1.1
...
user=philip&pw=123456



HTTP/1.1 200 OK
Set-Cookie: sessionid=3a4b5e
...

The user logs in by entering user name and password in an HTML form. The web app then creates a session in memory and returns a session ID.

GET home.php HTTP/1.1
Cookie: sessionid=3a4b5e
...

HTTP/1.1 200 OK
...

GET checkmail.php HTTP/1.1
Cookie: sessionid=3a4b5e
...

HTTP/1.1 200 OK
...

The browser attaches the cookie in all future requests to the web app. The web app can use the session id to look up application state (e.g. current user, 'session variables')

Cookie attributes

- A web server can restrict the scope of a cookie with attributes:
 - **expires** : date/time after which this cookie can be deleted
 - If not set, the cookie is deleted when user quits the browser
 - **path, domain** : the client should only include this cookie for requests in this domain and URL under this path
 - If not set, the default is the domain and path of the response
 - **secure**: a secure cookie may only be sent through SSL

```
HTTP/1.1 200 OK
```

```
Set-Cookie: userid=12345; expires=Fri, 31-Dec-2010  
23:59:59 GMT; path=/; domain=.example.com
```

```
...
```

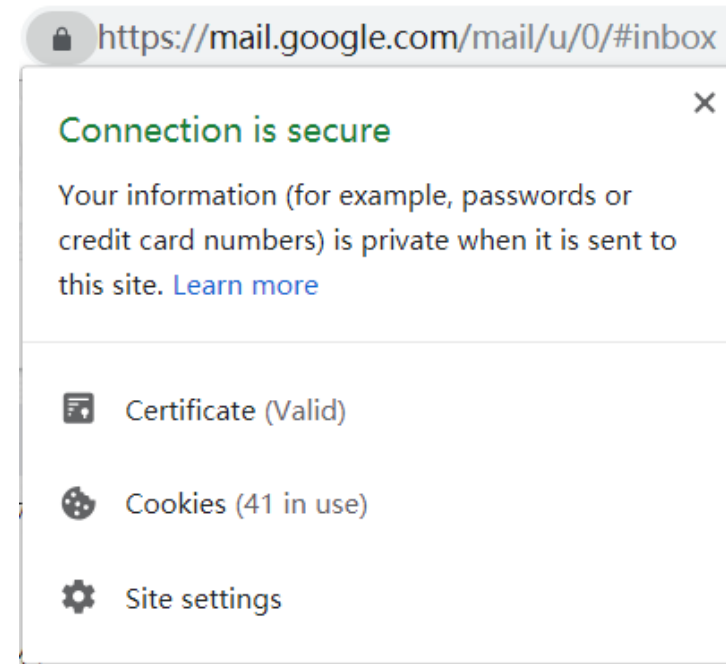
Encryption for HTTP

- **TLS (Transport Layer Security, aka. SSL)** are cryptographic protocols that encrypt data transmitted over a TCP connection.
 - Common versions: TLS1.2, TLS1.3
 - Can run different protocols over TLS, e.g. HTTP, SMTP, IMAP
- Two purposes:
 - Prevent eavesdropping and tampering
 - e.g. Only the client and server of an HTTP transaction can read the request/response
 - Verify the authenticity of the server
 - The server has a valid digital certificate issued by a certificate authority known by the browser

HTTPS

<https://mail.google.com/mail/#inbox>

- Need to install/trust a digital certificate in the web server
- https – runs HTTP over a secured TCP connection
 - Use port 443
 - Usually TLSv1.2 TLSv1.3
- A secure HTTP transaction
 - Attackers cannot read the request and response
 - Proxy (including cache servers) cannot read the messages either



Partially secure web page

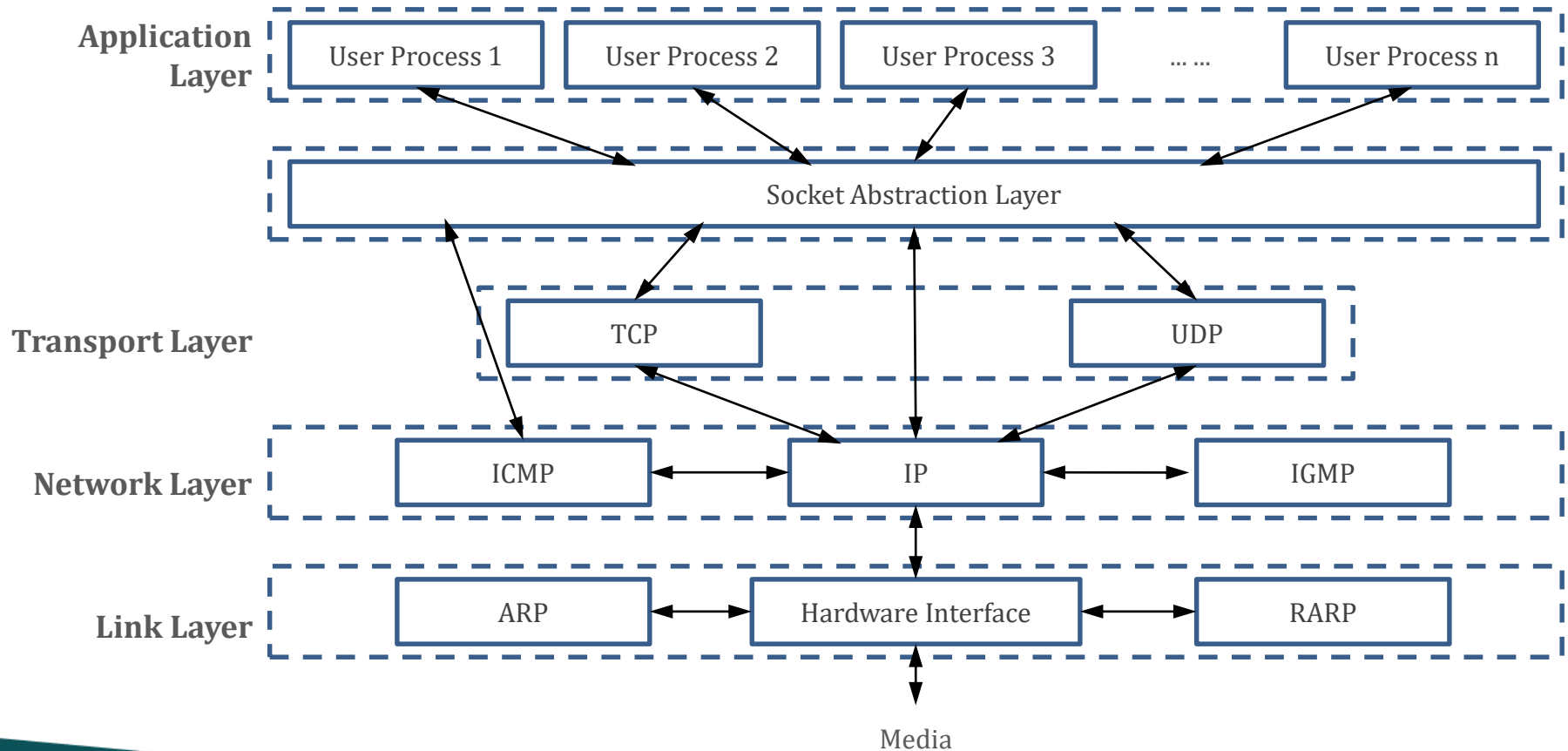
- A secure web page (https) that refers to unsecure resources (http)
 - e.g. the HTML page is using https, but the images inside are using http only
 - Unsecure resources may be modified and then added to the supposedly secure HTML page
 - Very serious if these are JavaScript files
 - HTTP requests to unsecure resources may contain cookies and eavesdropped by attackers
 - Problem solved by 'secure' attribute of cookies

✖ Mixed Content: The page at 'https://[redacted] index.html' was loaded over HTTPS, but requested an insecure resource 'http://player.[redacted]'. This request has been blocked; the content must be served over HTTPS.

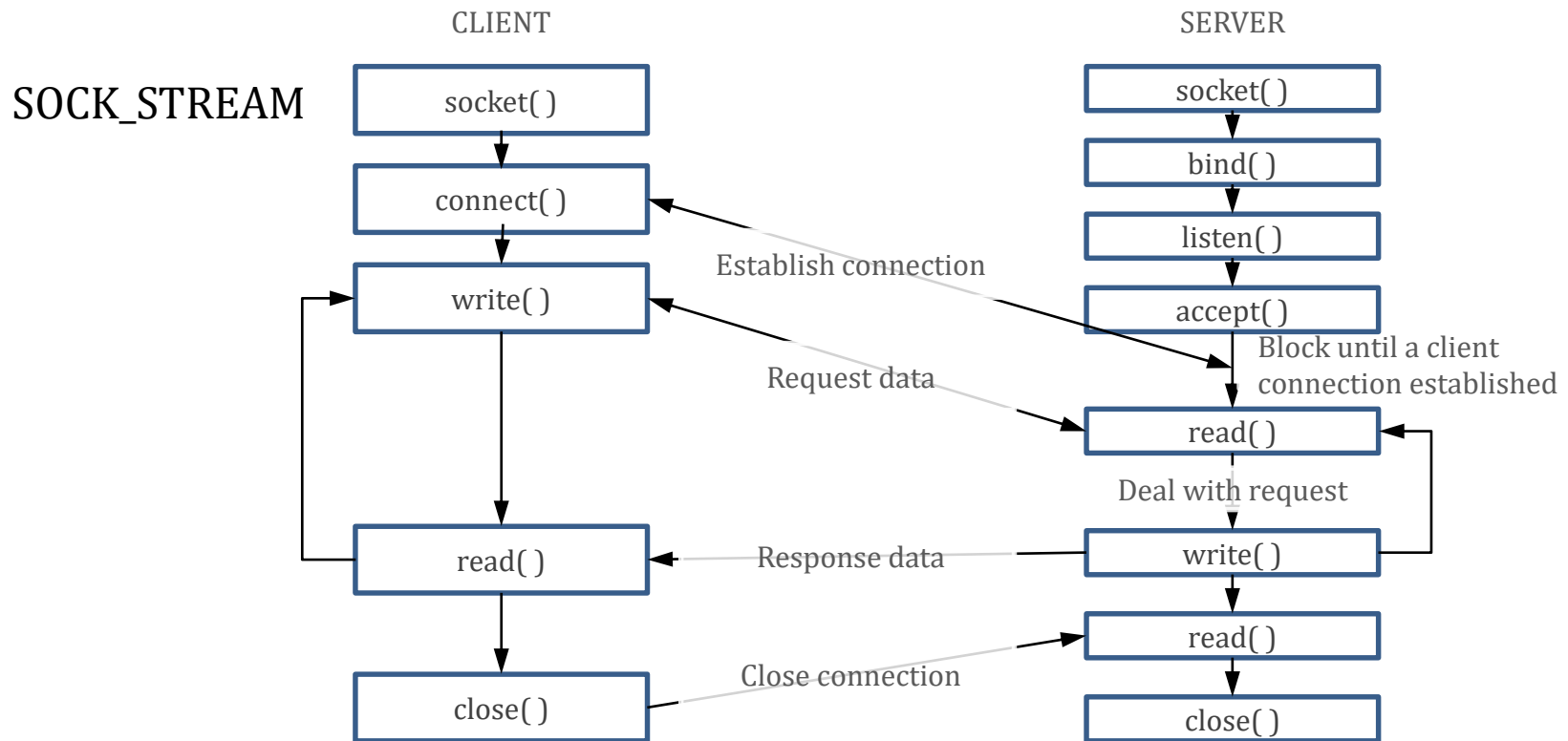
Part B.

Socket Programming

Socket (1)



Socket (2)



Socket Example 1: Echo Server (1)

```
import socket

def echo():
    sock = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
    sock.bind(('127.0.0.1', 5555))
    sock.listen(10)
    sock.settimeout(0.5)

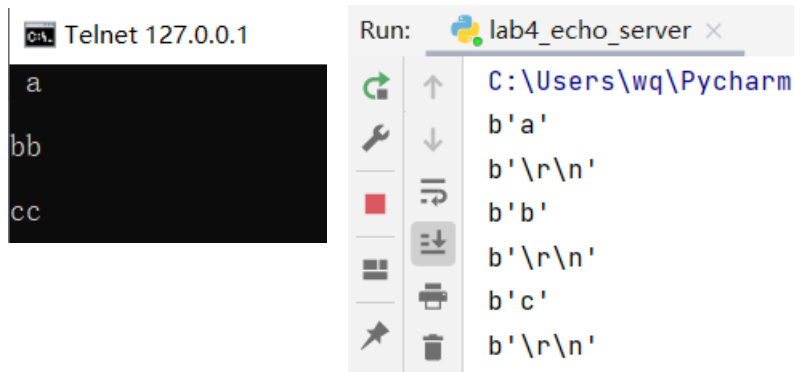
    while True:
        try:
            conn, address = sock.accept()
            while True:
                data = conn.recv(2048)
                if data and data != b'exit':
                    conn.send(data)
                    print(data)
                else:
                    conn.close()
                    break
            except socket.timeout:
                continue

if __name__ == "__main__":
    try:
        echo()
    except KeyboardInterrupt:
        pass
```

Socket Example 1: Echo Server (2)

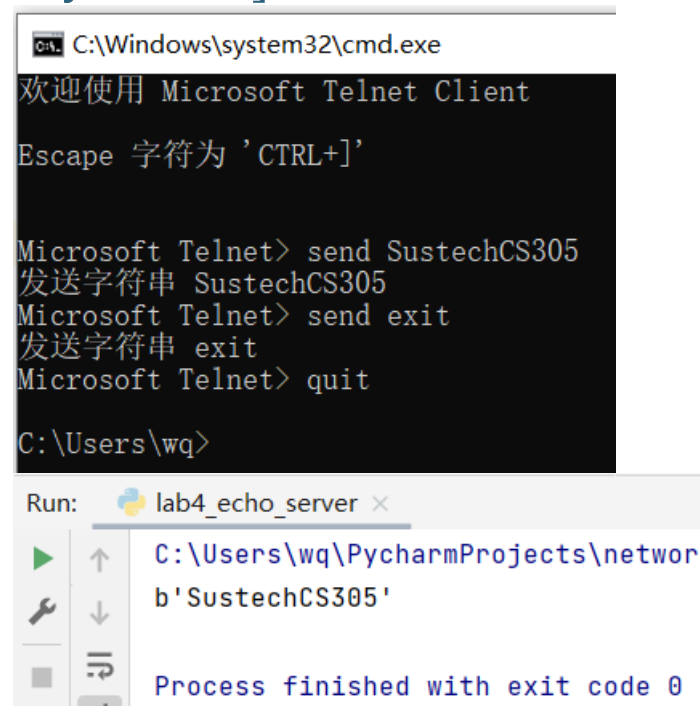
Running under Windows

```
C:\Users\wq>telnet 127.0.0.1 5555
```



The image shows two windows side-by-side. On the left is a Telnet client window titled 'Telnet 127.0.0.1' with a black background. It displays the input 'a' followed by 'bb' and 'cc' on separate lines. On the right is a PyCharm Run console window titled 'lab4_echo_server'. It shows the output of the server: 'b'a'', 'b'\r\n'', 'b'b'', 'b'\r\n'', 'b'c'', and 'b'\r\n''.

If we want to send a string instead of characters, shortcut keys “ctrl+]” can be used.



The image shows two windows side-by-side. On the left is a Microsoft Telnet Client window titled 'C:\Windows\system32\cmd.exe'. It displays the text: '欢迎使用 Microsoft Telnet Client', 'Escape 字符为 'CTRL+]', 'Microsoft Telnet> send SustechCS305', '发送字符串 SustechCS305', 'Microsoft Telnet> send exit', '发送字符串 exit', 'Microsoft Telnet> quit', and 'C:\Users\wq>'. On the right is a PyCharm Run console window titled 'lab4_echo_server'. It shows the output of the server: 'b'SustechCS305'' and 'Process finished with exit code 0'.

Tips: using command “quit” can exit sending mode.

Socket Example 1: Echo Server (3)

Running under other OS



```
/c/Users/light/PycharmProjects/CS305-2
light@DESKTOP-K4SPJVV MINGW64 /c/Users/light/PycharmProjects/CS305-2
$ python echo.py
b'test\r\n'
b'CS305 is Awsome.\r\n'

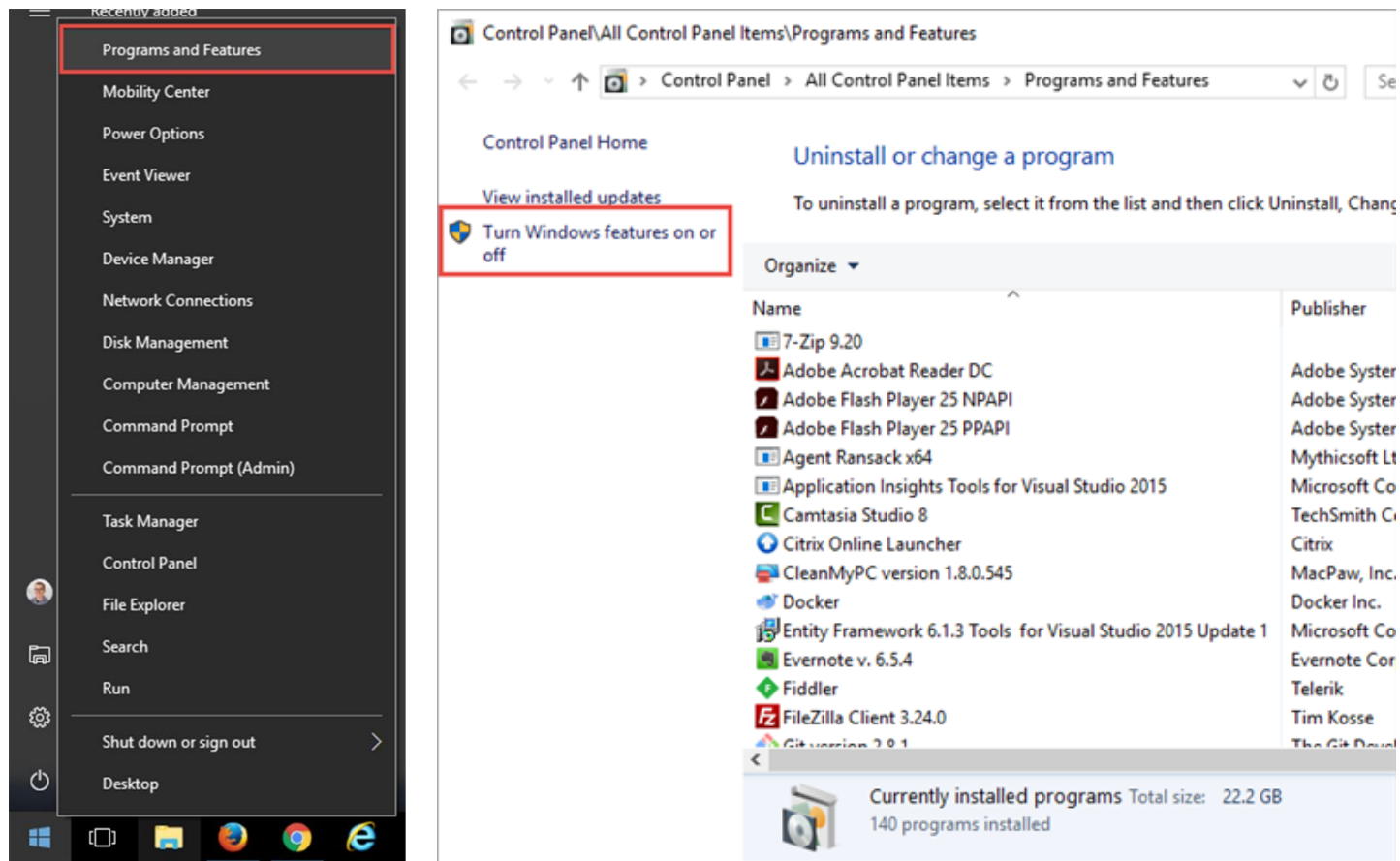
/
light@DESKTOP-K4SPJVV MINGW64 /
$ telnet 127.0.0.1 5555
Trying 127.0.0.1...
Connected to 127.0.0.1.
Escape character is '^]'.
test
test
CS305 is Awsome.
CS305 is Awsome.
exit
Connection closed by foreign host.

light@DESKTOP-K4SPJVV MINGW64 /
$
```

Practice 4.1 (Optional)

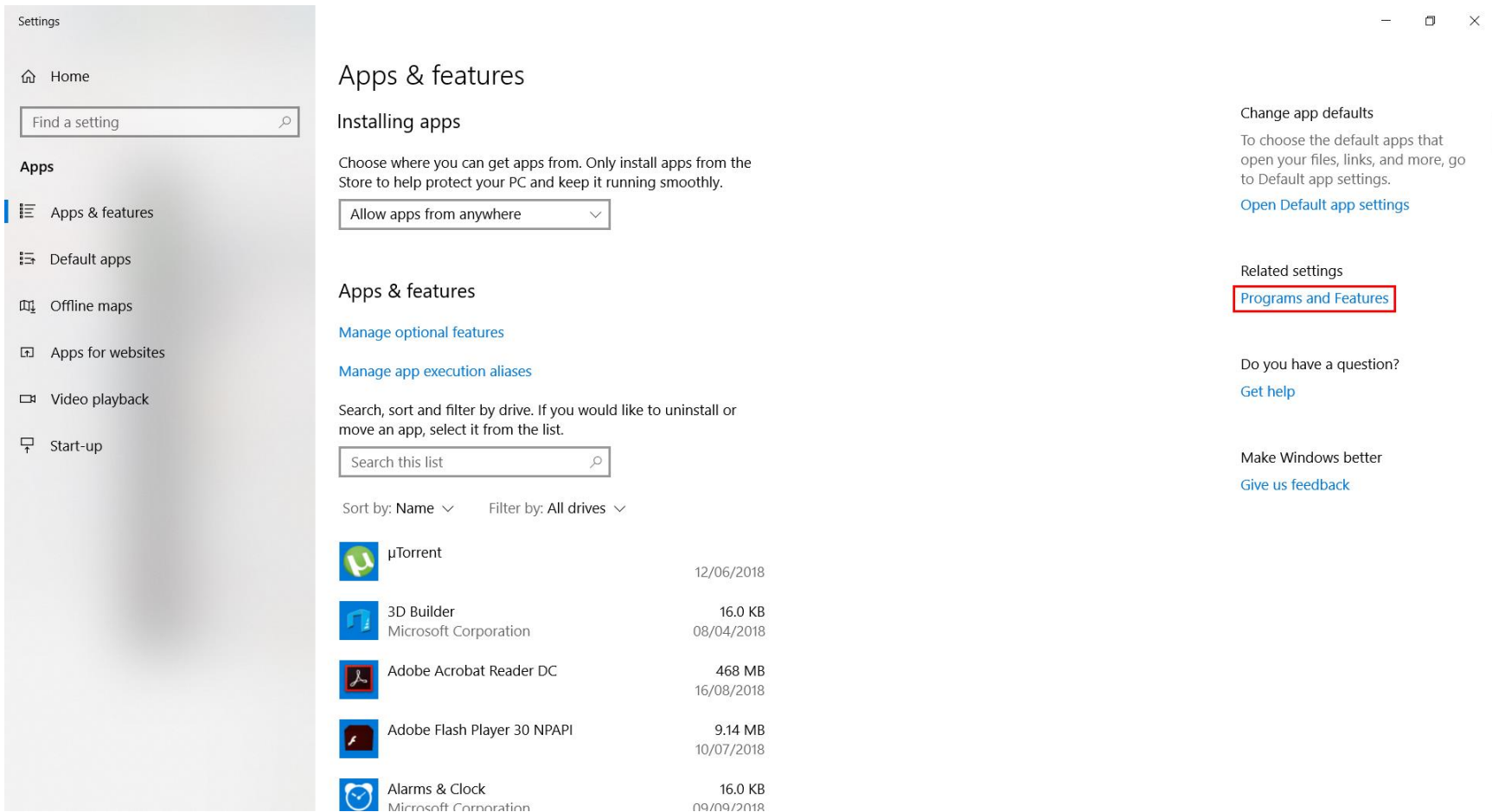
- 1. Run the server echo on Linux(or macOS) and Windows separately, is there any difference?
- 2. How to exit the loop? Can you design a method to quit elegantly?
- 3. Is there anyway to improve the server to make it work the same on different OS?

Tips 1: Enable Telnet on Windows (1)



Reference: <https://social.technet.microsoft.com/wiki/contents/articles/38433.windows-10-enabling-telnet-client.aspx>

Tips 1: Enable Telnet on Windows (2)



The screenshot shows the Windows Settings application, specifically the 'Apps & features' section. The left sidebar contains navigation options: Home, Find a setting, Apps, Apps & features (selected), Default apps, Offline maps, Apps for websites, Video playback, and Start-up. The main content area is titled 'Apps & features' and includes a sub-header 'Installing apps' with instructions on where to get apps from. A dropdown menu is set to 'Allow apps from anywhere'. Below this, there's a search bar and a list of installed applications. The list includes μTorrent, 3D Builder, Adobe Acrobat Reader DC, Adobe Flash Player 30 NPAPI, and Alarms & Clock. The right sidebar contains links for 'Change app defaults', 'Related settings' (with 'Programs and Features' highlighted), 'Do you have a question?', and 'Make Windows better'.

Settings

Home

Find a setting

Apps

- Apps & features
- Default apps
- Offline maps
- Apps for websites
- Video playback
- Start-up

Apps & features

Installing apps

Choose where you can get apps from. Only install apps from the Store to help protect your PC and keep it running smoothly.

Allow apps from anywhere

Apps & features

[Manage optional features](#)

[Manage app execution aliases](#)

Search, sort and filter by drive. If you would like to uninstall or move an app, select it from the list.

Search this list

Sort by: Name Filter by: All drives

	μTorrent	12/06/2018
	3D Builder	16.0 KB
	Microsoft Corporation	08/04/2018
	Adobe Acrobat Reader DC	468 MB
		16/08/2018
	Adobe Flash Player 30 NPAPI	9.14 MB
		10/07/2018
	Alarms & Clock	16.0 KB
	Microsoft Corporation	09/09/2018

Change app defaults

To choose the default apps that open your files, links, and more, go to Default app settings.

[Open Default app settings](#)

Related settings

[Programs and Features](#)

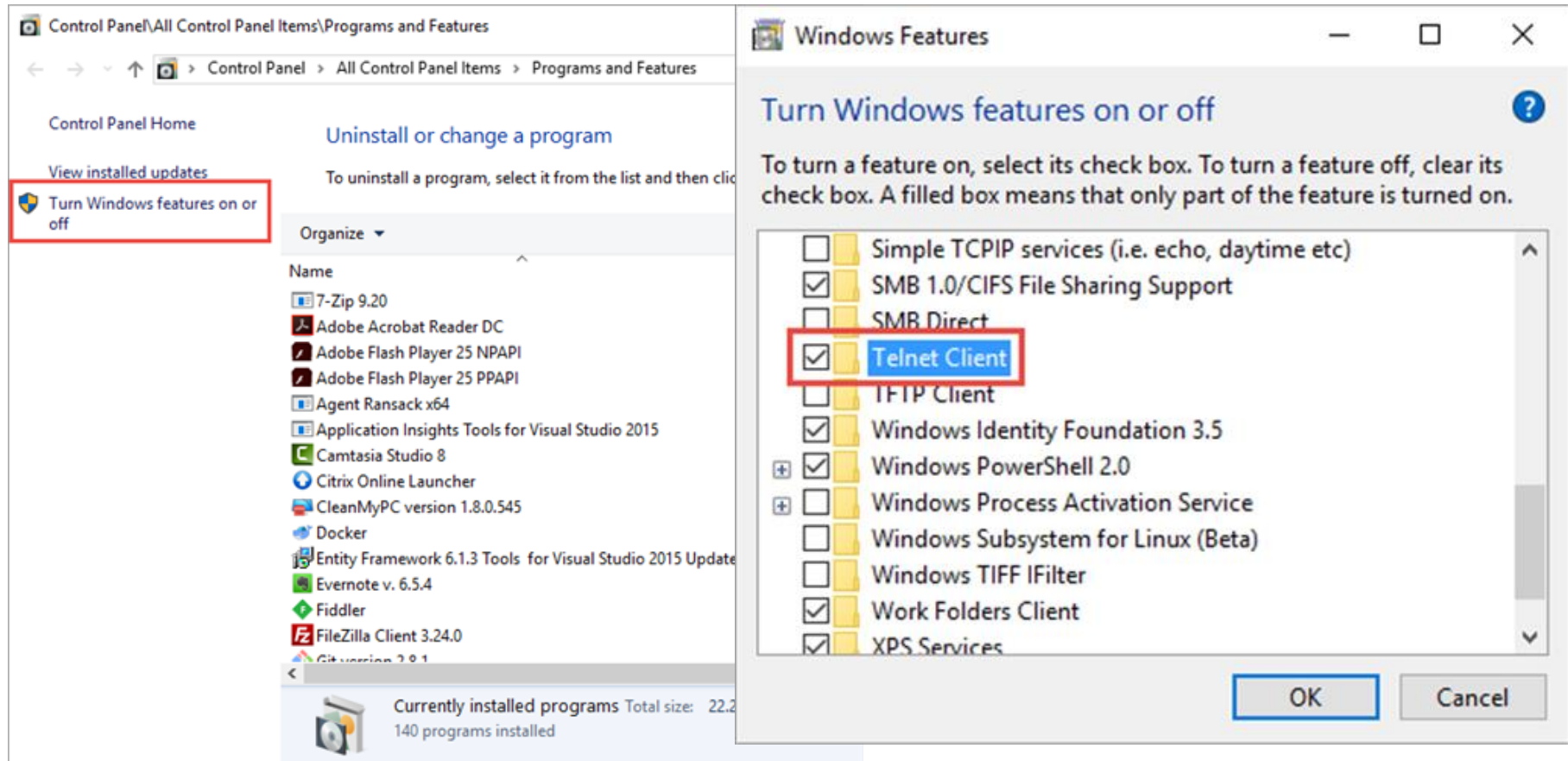
Do you have a question?

[Get help](#)

Make Windows better

[Give us feedback](#)

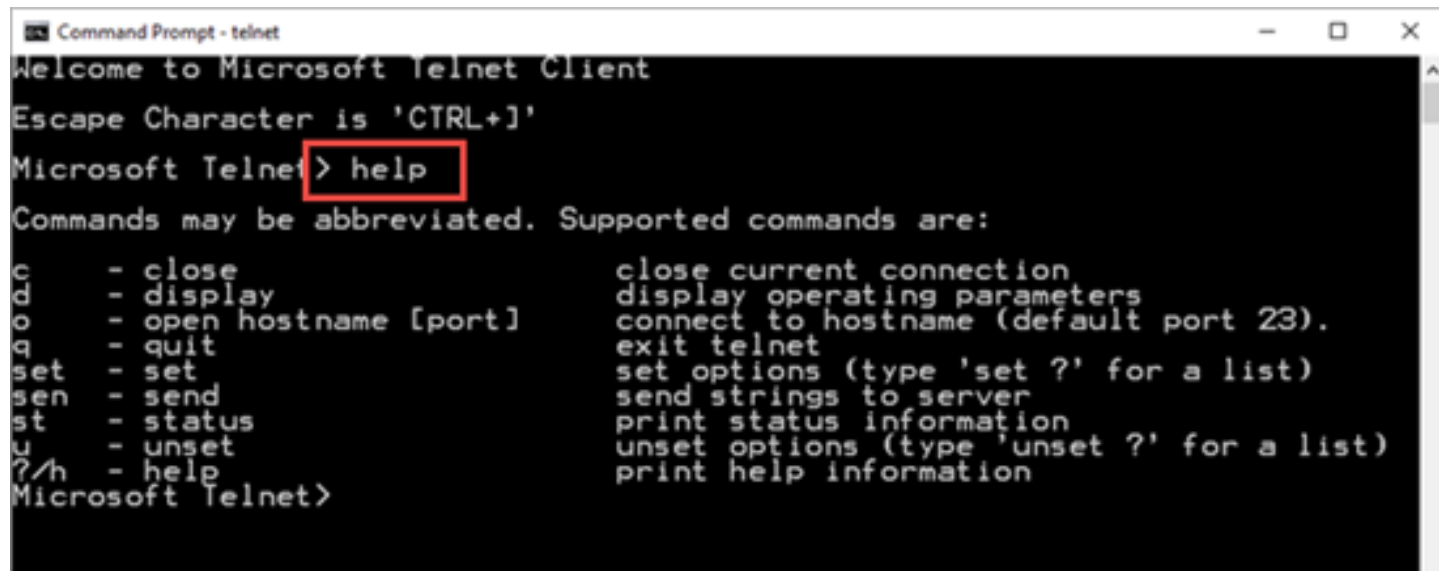
Tips 1: Enable Telnet on Windows (3)



Reference: <https://social.technet.microsoft.com/wiki/contents/articles/38433.windows-10-enabling-telnet-client.aspx>

Tips 1: Enable Telnet on Windows (4)

- Verify
 - Win+R, run “cmd”
 - Type “telnet”, press enter



```
Command Prompt - telnet
Welcome to Microsoft Telnet Client
Escape Character is 'CTRL+I'
Microsoft Telnet> help
Commands may be abbreviated. Supported commands are:
c      - close          close current connection
d      - display        display operating parameters
o      - open hostname [port] connect to hostname (default port 23).
q      - quit           exit telnet
set    - set            set options (type 'set ?' for a list)
sen    - send           send strings to server
st     - status         print status information
u      - unset          unset options (type 'unset ?' for a list)
?/h    - help           print help information
Microsoft Telnet>
```

Reference:

<https://social.technet.microsoft.com/wiki/contents/articles/38433.windows-10-enabling-telnet-client.aspx>

Example 2: Mimic a Simple Web Server (1)

```
def web():
```

```
    sock = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
```

```
    sock.bind(('127.0.0.1', 8080))
```

```
    sock.listen(10)
```

```
    while True:
```

```
        conn, address = sock.accept()
```

```
        data = conn.recv(2048).decode().split('\r\n')
```

```
        print(data[0].split(' '))
```

```
        res = err404
```

```
        if data[0].split(' ')[1] == '/':
```

```
            res = hello
```

```
        for line in res :
```

```
            conn.send(line)
```

```
        conn.close()
```

```
if __name__ == "__main__":
```

```
    try:
```

```
        web()
```

```
    except KeyboardInterrupt:
```

```
        pass
```

```
import socket
```

```
hello = [b'HTTP/1.0 200 OK\r\n',
```

```
        b'Connection: close'
```

```
        b'Content-Type:text/html; charset=utf-8\r\n',
```

```
        b'\r\n',
```

```
        b'<html><body>Hello World!<body></html>\r\n',
```

```
        b'\r\n']
```

```
err404 = [b'HTTP/1.0 404 Not Found\r\n',
```

```
        b'Connection: close'
```

```
        b'Content-Type:text/html; charset=utf-8\r\n',
```

```
        b'\r\n',
```

```
        b'<html><body>404 Not Found<body></html>\r\n',
```

```
        b'\r\n']
```

Example 2: Mimic a Simple Web Server (2)



The image shows two terminal windows. The top window is titled `/c/Users/light/PycharmProjects/CS305-2` and shows the execution of a Python script `web_hello.py`. The script receives two GET requests: one for `/` and one for `/not-exist`. The bottom window is titled `/` and shows the output of `curl` commands. The first `curl` command to `127.0.0.1:8080` returns `<html><body>Hello World!</body></html>`. The second `curl` command to `127.0.0.1:8080/not-exist` returns `<html><body>404 Not Found</body></html>`.

```
/c/Users/light/PycharmProjects/CS305-2
light@DESKTOP-K4SPJVV MINGW64 /c/Users/light/PycharmProjects/CS305-2
$ python web_hello.py
['GET', '/', 'HTTP/1.1']
['GET', '/not-exist', 'HTTP/1.1']

/
light@DESKTOP-K4SPJVV MINGW64 /
$ curl 127.0.0.1:8080
<html><body>Hello World!</body></html>

light@DESKTOP-K4SPJVV MINGW64 /
$ curl 127.0.0.1:8080/not-exist
<html><body>404 Not Found</body></html>
```

Example 3: Echo Server Multithreading (1)

```
import socket, threading
```

```
class Echo(threading.Thread):
```

```
    def __init__(self, conn, address):
```

```
        threading.Thread.__init__(self)
```

```
        self.conn = conn
```

```
        self.address = address
```

```
    def run(self):
```

```
        while True:
```

```
            data = self.conn.recv(2048)
```

```
            if data and data != b'exit':
```

```
                self.conn.send(data)
```

```
                print('{} sent: {}'.format(self.address, data))
```

```
            else:
```

```
                self.conn.close()
```

```
            return
```

```
def echo():
```

```
    sock = socket.socket(socket.AF_INET,  
                          socket.SOCK_STREAM)
```

```
    sock.bind(('127.0.0.1', 5555))
```

```
    sock.listen(10)
```

```
    while True:
```

```
        conn, address = sock.accept()
```

```
        Echo(conn, address).start()
```

```
if __name__ == "__main__":
```

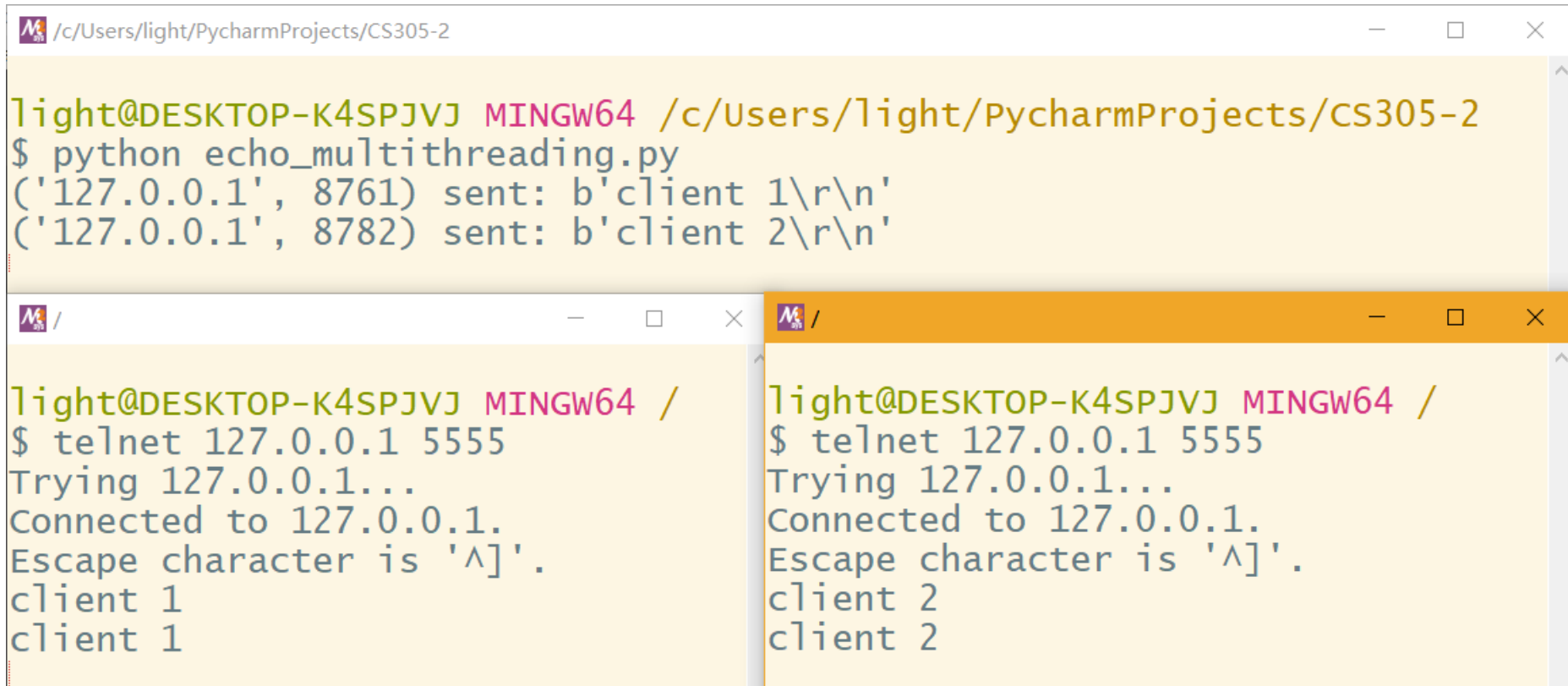
```
    try:
```

```
        echo()
```

```
    except KeyboardInterrupt:
```

```
        pass
```

Example 3: Echo Server Multithreading (2)



The image shows three terminal windows from a Windows command prompt. The top window shows the execution of a Python script named `echo_multithreading.py`. The script sends two messages to the console: `('127.0.0.1', 8761) sent: b'client 1\r\n'` and `('127.0.0.1', 8782) sent: b'client 2\r\n'`. The bottom-left window shows a telnet client connecting to `127.0.0.1` on port `5555`. It receives the message `client 1` from the server. The bottom-right window shows another telnet client connecting to `127.0.0.1` on port `5555`. It receives the message `client 2` from the server.

```
light@DESKTOP-K4SPJVV MINGW64 /c/Users/light/PycharmProjects/CS305-2
$ python echo_multithreading.py
('127.0.0.1', 8761) sent: b'client 1\r\n'
('127.0.0.1', 8782) sent: b'client 2\r\n'
```

```
light@DESKTOP-K4SPJVV MINGW64 /
$ telnet 127.0.0.1 5555
Trying 127.0.0.1...
Connected to 127.0.0.1.
Escape character is '^]'.
client 1
client 1
```

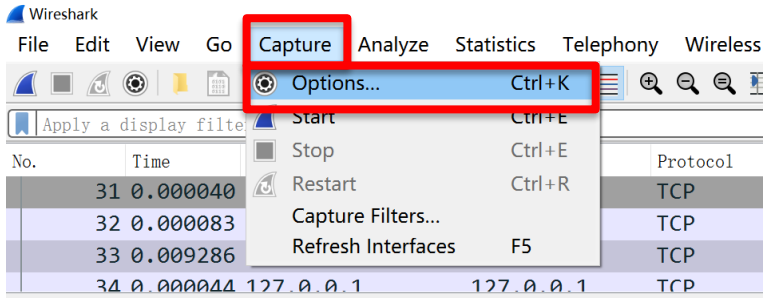
```
light@DESKTOP-K4SPJVV MINGW64 /
$ telnet 127.0.0.1 5555
Trying 127.0.0.1...
Connected to 127.0.0.1.
Escape character is '^]'.
client 2
client 2
```

Practice 4.2

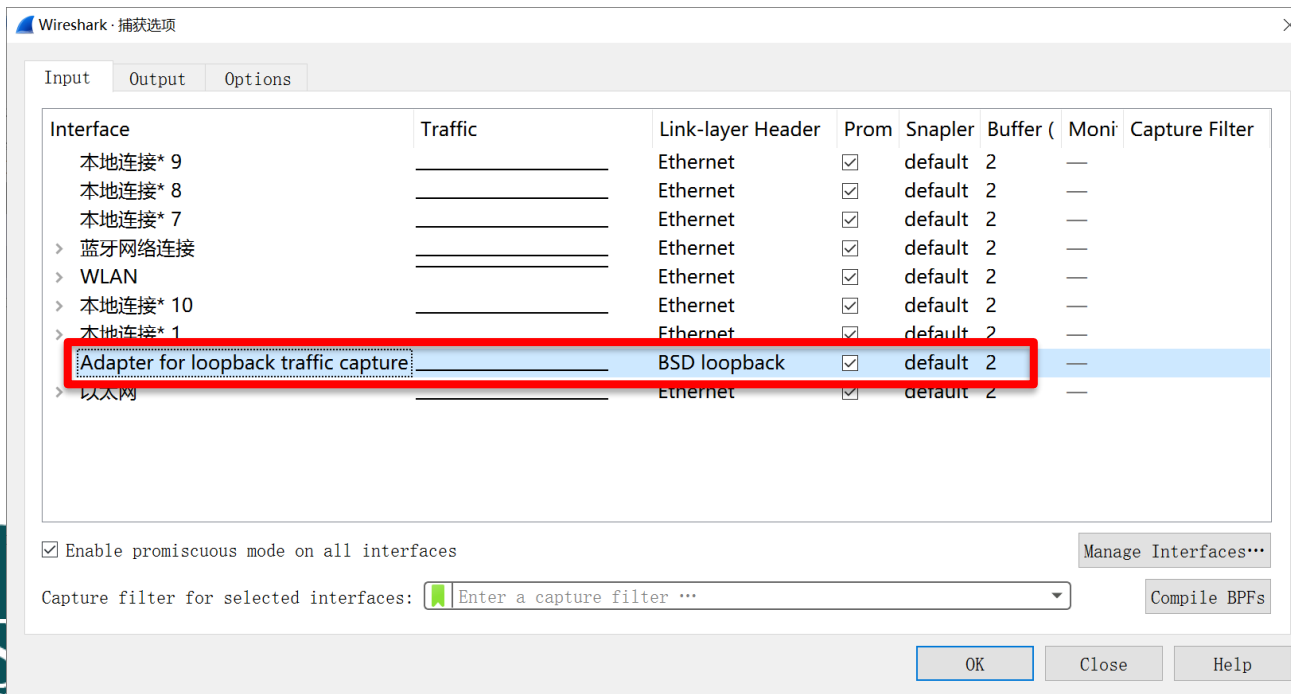
- Run example 2 and 3demo on your PC.
- Use Wireshark to capture and analyze the packets when running the demos. Find the corresponding message and list the source IPs, source port numbers, destination IPs, destination port numbers and status code of response messages in each connections.

	Src IP	Dst IP	Src port number	Dst port number	Status code
Request in Example 2					----
Response in Example 2					
Request in Example 3					----
Response in Example 3					

Tips 2: capture loopback traffic



No.	Time	Source	Destination	Protocol
16	0.000241	127.0.0.1	127.0.0.1	TCP
17	0.000022	127.0.0.1	127.0.0.1	TCP
18	0.000023	127.0.0.1	127.0.0.1	TCP
19	0.000012	127.0.0.1	127.0.0.1	TCP



Part C.

Configure your PC as an HTTP server

Using http.server(1)

- Run command “python -m http.server” on your PC to create the simplest HTTP server.
- Official documentation:
<https://docs.python.org/3/library/http.server.html>
- The server listens to port 8000 by default. The default can be overridden by passing the desired port number as an argument.
- By default, the server binds itself to all interfaces. The option -b/--bind specifies a specific address to which it should bind.
- By default, the server uses the current directory. The option -d/--directory specifies a directory to which it serves the files.

Using http.server(2)

- Server

```
C:\Users\wq>python -m http.server
Serving HTTP on :: port 8000 (http://[::]:8000/) ...
::ffff:10.25.70.41 - - [06/Mar/2023 10:03:31] "GET / HTTP/1.1" 200 -
::ffff:10.25.70.41 - - [06/Mar/2023 10:03:31] code 404, message File not found
::ffff:10.25.70.41 - - [06/Mar/2023 10:03:31] "GET /favicon.ico HTTP/1.1" 404 -
::ffff:10.25.70.41 - - [06/Mar/2023 10:03:46] "GET /test.txt HTTP/1.1" 200 -
::ffff:10.16.6.199 - - [06/Mar/2023 10:03:57] "GET / HTTP/1.1" 200 -
::ffff:10.16.6.199 - - [06/Mar/2023 10:04:08] "GET /Documents/ HTTP/1.1" 200 -
::ffff:10.16.6.199 - - [06/Mar/2023 10:04:11] code 404, message No permission to list directory
::ffff:10.16.6.199 - - [06/Mar/2023 10:04:11] "GET /Documents/My%20Pictures/ HTTP/1.1" 404 -
::ffff:10.16.6.199 - - [06/Mar/2023 10:04:22] code 404, message No permission to list directory
::ffff:10.16.6.199 - - [06/Mar/2023 10:04:22] "GET /Cookies/ HTTP/1.1" 404 -
::ffff:10.25. [REDACTED] - - [06/Mar/2023 10:04:39] "GET / HTTP/1.1" 200 -
::ffff:10.25. [REDACTED] - - [06/Mar/2023 10:04:44] "GET /Downloads/ HTTP/1.1" 200 -
```

Using http.server(3)

- Client 1(PC)



Directory listing for /

- [.android/](#)
- [.conda/](#)
- [.condarc](#)
- [.continuum/](#)
- [.m2/](#)
- [.packettracer](#)
- [.stm32cubeide/](#)
- [.stm32cubemx/](#)
- [.stmcube/](#)
- [.stmcufinder/](#)
- [.vscode/](#)
- [.Xilinx/](#)
- [1.txt](#)
- [3D Objects/](#)
- [AppData/](#)
- [Application Data/](#)

- Client 2 (mobile phone)

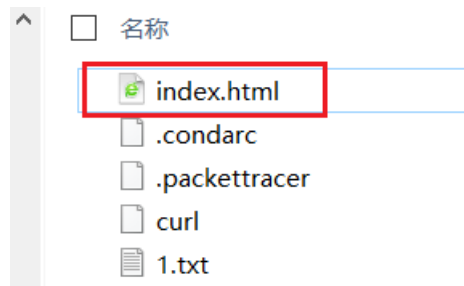
Directory listing for /

- [.android/](#)
- [.conda/](#)
- [.condarc](#)
- [.continuum/](#)
- [.m2/](#)
- [.packettracer](#)
- [.stm32cubeide/](#)
- [.stm32cubemx/](#)
- [.stmcube/](#)
- [.stmcufinder/](#)
- [.vscode/](#)
- [.Xilinx/](#)
- [1.txt](#)
- [3D Objects/](#)
- [AppData/](#)
- [Application Data/](#)
- [PycharmProjects/](#)
- [Recent/](#)
- [Saved Games/](#)



Using http.server(4)

- Add a file named “index.html” under the root of server.



File content:

```
<html>
This is a test demo of CS305
</html>
```

- Access the URL again.



Practice 4.3

- Suppose your IP address is `***.***.***.***`
- Run command “python -m http.server” on your PC.
- Use Wireshark to capture and analyze the packets.
 - 3-1. When accessing the web server from your own PC, which URL will work, “`***.***.***.***:8000`” or “`127.0.0.1:8000`” ?
 - 3-2. When accessing the web server from your own PC, which network adapter interface should you choose in Wireshark to capture the traffic?
 - 3-3. Let your classmate to access your web server, which URL will work, “`***.***.***.***:8000`” or “`127.0.0.1:8000`” ?
 - 3-4. Let your classmate to access your web server, which network adapter interface should you choose in Wireshark?
- Capture and analyze the packets, list the source IPs, source port numbers, destination IPs, destination port numbers and response’s status code of each connections.

Using flask(1)

- A lightweight web application framework written in Python.
- Official documentation:
<https://flask.palletsprojects.com/en/2.2.x/>
- Flask depends on the Jinja template engine and the Werkzeug WSGI toolkit.
- Jinja is a fast, expressive, extensible templating engine.
- Werkzeug is a comprehensive WSGI web application library.
(WSGI: Web Server Gateway Interface)

Using flask(2)

```
from flask import Flask
from flask import request
from flask import render_template

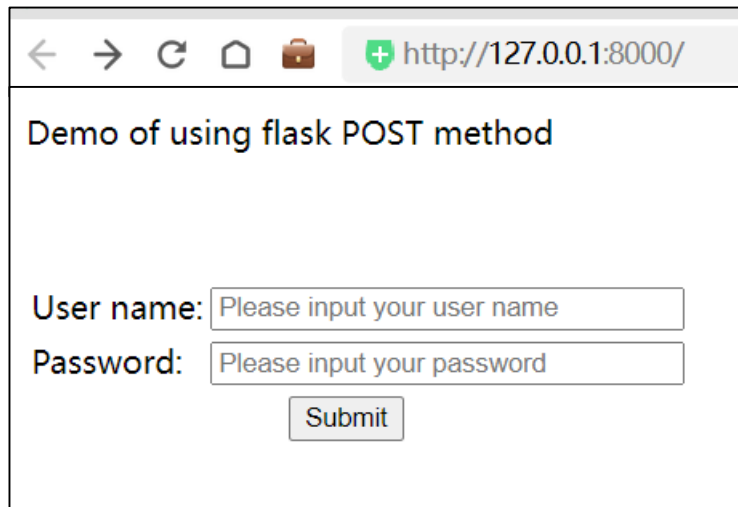
app = Flask(__name__)

@app.route("/", methods=['GET', 'POST'])
def main():
    if request.method == 'GET':
        return render_template('main.html')
    if request.method == 'POST':
        user = request.form.get('user')
        password = request.form.get('password')
        res = "User: %s\nPassword: %s" % (user, password)
        return res

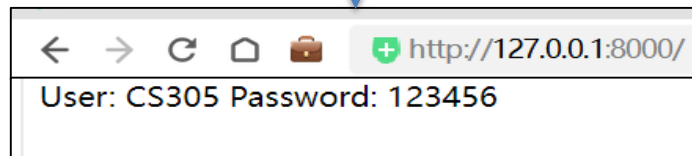
if __name__ == "__main__":
    app.run(host='0.0.0.0', port=7000, debug=True)
```

Using flask(3)

- Client 1

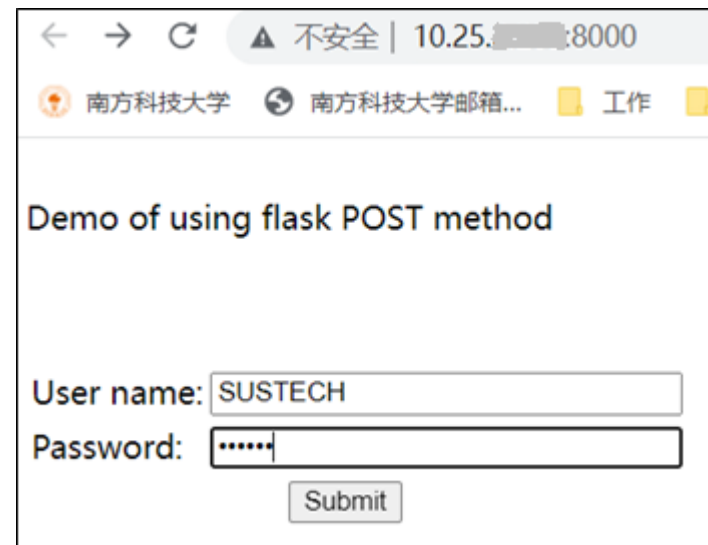


A screenshot of a web browser window. The address bar shows 'http://127.0.0.1:8000/'. The page title is 'Demo of using flask POST method'. Below the title, there are two input fields: 'User name:' with placeholder text 'Please input your user name' and 'Password:' with placeholder text 'Please input your password'. A 'Submit' button is located below the password field.



A screenshot of the browser window after submission. The address bar remains the same. The page content now displays 'User: CS305 Password: 123456'.

- Client 2



A screenshot of a web browser window. The address bar shows '10.25.8000' with a warning icon and the text '不安全'. The page title is 'Demo of using flask POST method'. Below the title, there are two input fields: 'User name:' with the value 'SUSTECH' and 'Password:' with masked characters '.....'. A 'Submit' button is located below the password field.



A screenshot of the browser window after submission. The address bar remains the same. The page content now displays 'User: SUSTECH Password: 654321'.

Tips 3:

- Flask uses port 5000 by default, you can choose other port numbers when running as demo shows.
- When running the demo, you should put main.html in a folder named “templates”. For example, suppose the python project locates under “D:\CS305\http”, and then you should put main.html in “D:\CS305\http\templates”.

Practice 4.4(optional)

- Suppose your IP address is `***.***.***.***`
- Run the demo using flask on your PC.
- Use Wireshark to capture and analyze the packets.
 - 4-1. When accessing the web server from your own PC, which URL will work, `***.***.***.***:7000` or `127.0.0.1:7000` ? Which network adapter interface should you choose in Wireshark to capture the traffic?
 - 4-2. Let your classmate to access your web server, which URL will work, `***.***.***.***:7000` or `127.0.0.1:7000` ? Which network adapter interface should you choose in Wireshark?
- If we need analyze the packages that using POST method, what display filter should be set? Are the packages request messages or response messages?

351	0.000340	10.25.2.205	10.25.2.205	HTTP	667	POST / HTTP/1.1 (application/x-www-form-urlencoded)
454	0.000267	127.0.0.1	127.0.0.1	HTTP	868	POST / HTTP/1.1 (application/x-www-form-urlencoded)